USN | | | | | | | | | | [23MSCH1H03]

FIRST SEMESTER UNIVERSITY EXAMINATION, JANUARY - 2025

PHYSICAL CHEMISTRY - I

Time: 03 HOURS **Maximum Marks:50**

		Blooms Level	CO	Marks
	SECTION – A			
1.	**Answer any FOUR of the following questions.**			**4×5=20**
a.	**i.** Justify Gibbs Helmholtz and Richard concept on heat theorem. **ii.** Explain Maxwell relation with respect to Gibbs free energy.	1	1	**05**
b.	**i.** How does fugacity vary with pressure? **ii.** Derive the Clapeyron equation.	4	2	**05**
c.	What are the main features of photoelectric effect? How did Einstein explain the phenomenon?	3	3	**05**
d.	Explain the methods to find out the rate of reaction.	2	4	**05**
e.	Illustrate briefly about Debye-Huckel limiting law.	3	5	**05**
	SECTION – B			
2.	**Answer any TWO of the following questions.**			**2×9=18**
a.	Determine the effect of temperature and pressure on fugacity.	4	2	**09**
b.	What is steady state approximation? Find the rate of any photochemical reaction using steady-state approximation.	3	4	**09**
c.	Explain the equation for relaxation time and its significance in the relaxation method.	2	6	**09**
	SECTION – C			
3.	**Answer the following question.**			**1x12=12**
a.	**i.** Illustrate the proof of the Nernst heat theorem.	1	1	**04**
	ii. Calculate the maximum kinetic energy of photoelectron (in eV) emitted on shining light of wavelength 6.2×10^{-6} m on a metal surface. The work function of the metal is 0.1 eV.	5	3	**04**
	iii. Illustrate with schematic digram potential-distance plot of Gouy-Chapmann and Stern model.	4	5	**04**